

Business Requirements Document (BRD)

QADemo E-Commerce Application

Document Type: Business Requirements Document
Project: QADemo E-Commerce Application
Version: 1.0
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Executive Summary

The **QADemo E-Commerce Application** is an online shopping platform designed to allow customers to browse products, manage their shopping cart, place orders and complete purchases through a convenient digital interface.

The purpose of this project is to provide a realistic e-commerce environment that supports the complete customer purchasing journey, from product discovery through order completion. The application should provide a reliable and user-friendly shopping experience while enabling the business to manage products, customers, orders and sales activities efficiently.

The project will also provide a suitable environment for validating critical e-commerce business processes and ensuring that the application meets defined business expectations.

Business Objectives

The primary business objectives of the QADemo E-Commerce Application are:

- Provide customers with a convenient platform for browsing and purchasing products online.
- Allow customers to search for and discover products efficiently.
- Enable customers to manage selected products through a shopping cart.
- Provide a complete and reliable online checkout and ordering process.
- Support secure and convenient customer transactions.
- Provide customers with visibility into their orders.
- Improve the overall digital shopping experience.
- Ensure that important e-commerce business processes operate consistently and reliably.
- Reduce potential business loss caused by incorrect product, cart, order or transaction processing.
- Provide a platform that can be used to evaluate and improve the quality of an e-commerce application.

Scope

1. In-Scope

The following business capabilities are within the scope of the QADemo E-Commerce Application:

Customer Account Management

- Customer registration.
- Customer login and logout.
- Customer account management.
- Customer profile information management.

Product Management and Discovery

- Display available products.
- Browse products by available categories.
- Search for products.
- View product details.
- View product availability and relevant product information.

Shopping Cart

- Add products to the shopping cart.
- Remove products from the shopping cart.
- Update product quantities.
- View selected products and their associated prices.
- Review the cart before placing an order.

Checkout and Order Placement

- Proceed from the shopping cart to checkout.
- Provide required customer and delivery information.
- Review order information before confirmation.
- Place an order.
- Receive confirmation of a successfully placed order.

Order Management

- View order information.
- Track the status of an order where supported.
- Review previously placed orders where supported.

Payment

- Support the payment process required to complete an order.
- Provide appropriate transaction results to the customer.

2. Out-of-Scope

The following capabilities are outside the scope of the initial project unless specifically required later:

- Social media login.
- Advanced loyalty and rewards programs.
- Subscription-based purchases.
- International shipping management.
- Multi-vendor marketplace functionality.
- Advanced recommendation and personalization systems.
- AI-based customer support.
- Cryptocurrency payments.
- Complex business analytics and forecasting.
- Physical warehouse management.
- Supplier and procurement management.

Stakeholders

Stakeholder	Responsibility / Interest
Business Owner	Defines business objectives and evaluates overall business value.
Product Manager	Defines product vision, priorities and customer needs.
Business Analyst	Gathers, analyzes, documents and validates business requirements.
Development Team	Builds and maintains the application according to approved requirements.
QA/SQA Team	Validates that the application satisfies business and functional requirements.
UI/UX Team	Designs a convenient and understandable customer experience.
Customer	Browses products, purchases products, and manages orders.
Administrator	Manages relevant products, customers and orders according to business permissions.

Business Requirements

BR-001 — Customer Authentication

The business requires registered customers to securely access their accounts so that customer-specific information and shopping activities can be protected.

Priority: Critical

Stakeholder: Customer / Business Owner

BR-002 — Product Discovery

The business requires customers to be able to browse available products so that they can discover products of interest.

Priority: Critical

Stakeholder: Customer / Product Manager

BR-003 — Product Search

The business requires customers to be able to search for available products so that they can efficiently find products of interest.

Priority: Critical

Stakeholder: Customer / Product Manager

BR-004 — Product Information

The business requires customers to be able to view relevant product information before making a purchasing decision.

Priority: High

Stakeholder: Customer / Product Manager

BR-005 — Shopping Cart

The business requires customers to be able to add products to a shopping cart and manage their selected products before checkout.

Priority: Critical

Stakeholder: Customer / Business Owner

BR-006 — Order Calculation

The business requires the application to accurately calculate the total value of selected products before an order is placed.

Priority: Critical

Stakeholder: Customer / Business Owner

BR-007 — Checkout

The business requires customers to be able to provide the information necessary to complete an order and review their purchase before final confirmation.

Priority: Critical

Stakeholder: Customer / Business Owner

BR-008 — Order Placement

The business requires customers to be able to successfully place an order for eligible products through the application.

Priority: Critical

Stakeholder: Customer / Business Owner

BR-009 — Payment

The business requires the application to support the required payment process so that customers can complete their purchases.

Priority: Critical

Stakeholder: Customer / Business Owner

BR-010 — Order Confirmation

The business requires customers to receive confirmation after successfully placing an order so that they have evidence that their purchase has been completed.

Priority: High

Stakeholder: Customer

BR-011 — Product Administration

The business requires authorized administrators to manage relevant products so that e-commerce operations can be maintained effectively.

Priority: High

Stakeholder: Administrator / Business Owner

BR-012 — Order Administration

The business requires authorized administrators to manage customer orders so that e-commerce operations can be maintained effectively.

Priority: High

Stakeholder: Administrator / Business Owner

Business Rules

The following business rules apply to the e-commerce process:

- Customers should only be able to place orders for products that are eligible for purchase.
- Product prices displayed during checkout should accurately reflect the applicable product prices.
- An order should not be considered successfully placed until the required order process has been completed.
- Customers should receive appropriate confirmation after a successful order.
- Only authorized users should have access to administrative functionality.
- Customer account and order information should be appropriately protected.
- The application should maintain accurate information throughout the shopping and ordering process.

Constraints

The project may be subject to the following constraints:

- The application must operate within the capabilities of the existing QADemo environment.
- The project scope may be limited to the functionality available in the application.
- External payment, email or third-party services may impose availability or integration limitations.
- Development and testing activities may be constrained by project time and available resources.
- Business requirements may change based on stakeholder feedback.

Assumptions

The following assumptions are made for the project:

- Customers have access to a compatible device and internet connection.
- Required product information is available to the application.
- The application has the necessary infrastructure to support customer shopping activities.
- Required external services are available when needed.
- Stakeholders will provide clarification when business requirements are unclear.
- Product, customer and order information provided by the system is expected to be sufficiently accurate for business operations.

Business Risks

Risk	Potential Business Impact	Mitigation
Incorrect product information	Customers may make incorrect purchasing decisions.	Validate product information and related business processes.
Incorrect cart/order calculation	Financial loss and customer dissatisfaction.	Thoroughly validate pricing and calculation processes.
Checkout failure	Customers may abandon purchases.	Test the complete checkout workflow.
Payment failure	Lost sales and customer dissatisfaction.	Validate successful and unsuccessful payment scenarios.
Unauthorized access	Customer/business data may be exposed.	Validate authentication and authorization controls.
Order processing errors	Incorrect or lost orders.	Test order creation and order-status processes thoroughly.
Poor usability	Increased customer abandonment.	Validate important customer journeys from a usability perspective.

Business Benefits

Successful implementation of the QADemo E-Commerce Application is expected to provide the following benefits:

- Convenient online shopping for customers.
- Improved accessibility to products.
- Faster and more structured purchasing processes.
- Better management of customer orders.
- Reduced errors in manual purchasing activities.
- Improved customer experience.
- Better visibility into the overall customer purchasing journey.
- A reliable platform for demonstrating and evaluating e-commerce business processes.

Success Criteria

The project will be considered successful when:

- Customers can successfully access their accounts.
- Customers can discover and view available products.
- Customers can add and manage products in their shopping carts.
- Customers can proceed through checkout successfully.
- Customers can place eligible orders successfully.

- Customers receive appropriate confirmation after successful orders.
- Administrative users can perform their authorized business activities.
- Critical business workflows operate reliably without major business-impacting defects.
- The application provides a consistent and satisfactory customer shopping experience.

Dependencies

The project may depend on:

- Product and inventory information.
- Customer account information.
- Order management functionality.
- Payment services or payment simulation mechanisms.
- Email/notification services where applicable.
- Application infrastructure and hosting environment.
- Availability of development, QA and business stakeholders.

Requirement Priority

Business requirements will generally be prioritized as follows:

Critical: Essential for completing the primary business transaction.

High: Important for normal customer or business operations.

Medium: Valuable functionality that supports the overall product but does not prevent the primary transaction.

Low: Nice-to-have functionality that can be considered for future releases.

Approval

Role	Name	Approval	Date
Business Owner	_____	_____	_____
Product Manager	_____	_____	_____
Business Analyst	_____	_____	_____
QA/SQA Representative	_____	_____	_____

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